

A REPORT
ON
FINAL PROJECT
Design & Analysis of 6T SRAM CELL

Submitted by

K. Eshwari

AP23110020113

Department of ECE

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Under The Supervision Of

Dr. Bharat Bhushan Upadhyay

Department of Electronics & Communication Engineering

SRM UNIVERSITY - AP



Department of Electronics & Communication Engineering

SRM University-AP

Neerukonda

Mangalgiri-522503

Andhra Pradesh

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1. Abstract

Static Random Access Memory (SRAM) is a critical component in modern high-performance digital systems, primarily due to its superior speed, low latency, and robustness compared to other memory technologies. This report focuses on the detailed design and analysis of a conventional 6-Transistor (6T) SRAM cell implemented using CMOS technology. The 6T SRAM cell is widely adopted in cache memory and embedded systems due to its balance between stability, speed, and area efficiency.

In this work, the SRAM cell is designed and simulated using industry-standard Electronic Design Automation (EDA) tools such as Cadence Virtuoso and Spectre simulator. The design methodology includes careful transistor sizing, schematic implementation, and testbench configuration to evaluate performance under realistic conditions. The study emphasizes both static and dynamic characteristics of the SRAM cell.

The analysis includes DC characterization through Voltage Transfer Characteristics (VTC) and Static Noise Margin (SNM) evaluation using the butterfly curve method, which is a key indicator of cell stability. Additionally, transient analysis is performed to observe the behavior of the SRAM cell during read and write operations, including delay and switching characteristics.

The results demonstrate the trade-offs involved in SRAM design, particularly between read stability, write ability, and power consumption. The findings highlight that proper transistor sizing is essential to ensure reliable operation, especially in scaled technologies where process variations and leakage currents significantly impact performance.

2. Introduction

In modern digital systems, memory plays a vital role in determining overall system performance. With the increasing demand for high-speed computation in applications such as microprocessors, graphics processing units (GPUs), and system-on-chip (SoC) designs, the need for efficient and fast memory has become more significant than ever. Among the different types of semiconductor memories, Static Random Access Memory (SRAM) is widely used due to its fast access time and reliability.

SRAM is primarily utilized in cache memory levels (L1, L2, L3) of processors, where speed is more critical than storage density. Unlike Dynamic Random Access Memory (DRAM), SRAM does not require periodic refreshing, which simplifies its operation and significantly reduces access latency. However, this advantage comes at the cost of increased area since SRAM cells use more transistors per bit.

The most commonly used SRAM cell configuration is the 6T SRAM cell, which consists of six transistors forming a bistable latch capable of storing one bit of information. This configuration provides a good compromise between stability, power consumption, and silicon area, making it a standard choice in VLSI design.

The design and analysis of SRAM cells have become increasingly challenging with technology scaling into deep submicron regions. Issues such as reduced supply voltage, increased leakage currents, and process variations affect the stability and reliability of SRAM cells. Therefore, understanding the internal operation, design constraints, and performance metrics of SRAM cells is essential for designing robust memory systems.

This report aims to provide a comprehensive study of the 6T SRAM cell, covering its architecture, modes of operation, design methodology, and detailed analysis using simulation techniques. The objective is to evaluate its performance in terms of stability, speed, and power consumption while identifying the key challenges associated with its design.

3. Methodology

The design of the 6T SRAM cell was carried out using a systematic and structured methodology to ensure optimal performance in terms of stability, speed, and power consumption. The complete design flow includes technology selection, transistor sizing, schematic design, and simulation setup using industry-standard Electronic Design Automation (EDA) tools.

The first step in the design process involves selecting an appropriate CMOS technology node. In this work, a standard deep-submicron technology such as 45 nm or 90 nm is considered, which reflects practical design scenarios used in modern VLSI systems. The choice of technology node significantly impacts parameters such as threshold voltage, leakage current, and overall device performance. Lower technology nodes offer higher speed but introduce challenges such as increased leakage and variability.

A critical aspect of SRAM design is transistor sizing. The 6T SRAM cell consists of six transistors: two pull-up PMOS transistors, two pull-down NMOS transistors, and two access NMOS transistors. The relative strength of these transistors determines the stability and functionality of the cell. The pull-down NMOS transistors are designed to be stronger than the access transistors to ensure that the stored data is not disturbed during the read operation. Similarly, the access transistors must be stronger than the pull-up PMOS transistors to allow successful write operations. This relationship is typically expressed using ratios such as the cell ratio (pull-down to access transistor strength) and pull-up ratio, which are carefully optimized during the design.

Following transistor sizing, the schematic of the 6T SRAM cell is implemented using Cadence Virtuoso. The design includes proper connections between the cross-coupled inverters and access transistors, along with the inclusion of bitlines (BL and $\overline{BL}$) and the word line (WL). The internal storage nodes are labeled as Q and QB, representing complementary logic states. Special attention is given to ensure symmetry in the layout to minimize mismatch and improve stability.

Once the schematic is completed, a testbench is created to simulate the SRAM cell under different operating conditions. The testbench includes voltage sources for supply (VDD), control signals for the word line, and input signals for the bitlines. Separate simulation setups are prepared for analyzing hold, read, and write operations. Pulse generators are used to activate the word line and control the timing of read and write cycles.

The design is then simulated using the Spectre simulator to evaluate both DC and transient characteristics. DC analysis is performed to obtain the voltage transfer characteristics and to generate the butterfly curve for Static Noise Margin (SNM) calculation. Transient analysis is conducted to observe the dynamic behavior of the cell during switching operations, including read and write delays.

4. 6T SRAM Cell Architecture

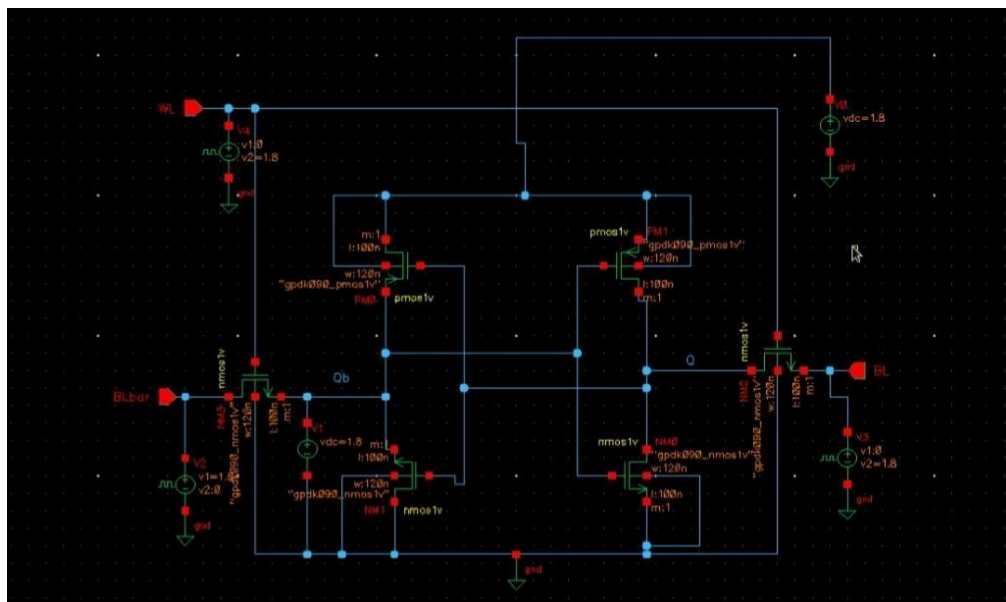
The 6T SRAM cell is composed of six transistors arranged to form a stable memory element capable of storing one bit of data. The core of the cell consists of two cross-coupled CMOS inverters, which act as a bistable latch. These inverters are formed by two PMOS pull-up transistors and two NMOS pull-down transistors.

In addition to the inverter pair, two NMOS access transistors are used to connect the internal storage nodes to the bitlines (BL and $\overline{BL}$). These access transistors are controlled by a word line (WL), which determines whether the cell is connected to the external circuitry.

The internal nodes of the SRAM cell are typically labeled as Q and QB, representing the stored value and its complement, respectively. When Q is high, QB is low, and vice versa. This complementary structure ensures strong feedback, which helps maintain data stability.

The operation of the 6T SRAM cell depends heavily on transistor sizing. The pull-down NMOS transistors must be strong enough to hold the stored value during a read operation, while the access transistors must be strong enough to allow data to be written into the cell. Similarly, the pull-up PMOS transistors must provide sufficient strength to maintain the stored value without excessive power consumption.

The design of the 6T SRAM cell requires careful balancing of these transistor strengths to achieve optimal performance. Improper sizing can lead to issues such as read disturb, write failure, or reduced noise margin.



5. Modes of Operation

The operation of a 6T SRAM cell can be classified into three primary modes: hold, read, and write. Each mode involves different circuit conditions and presents unique challenges.

Hold Mode

In hold mode, the word line is disabled ($WL = 0$), which turns off the access transistors and isolates the cell from the bitlines. The cross-coupled inverters maintain the stored value through positive feedback. This mode is critical for data retention and must be stable against leakage currents and noise.

Read Operation

During the read operation, both bitlines are typically precharged to a high voltage (V_{DD}). When the word line is enabled, the access transistors connect the internal nodes to the bitlines. Depending on the stored value, one of the bitlines begins to discharge through the pull-down transistor.

A major challenge during the read operation is the read disturb problem, where the act of reading can unintentionally alter the stored data. This occurs if the voltage at the internal node rises above a critical threshold due to voltage division between transistors.

Write Operation

In the write operation, the bitlines are driven with the desired data and its complement. When the word line is enabled, the strong drivers connected to the bitlines force the internal nodes to switch states. The new data overwrites the existing value in the cell.

Write operation requires the access transistors to be strong enough to overpower the feedback of the cross-coupled inverters. If the pull-up transistors are too strong, write failure may occur.

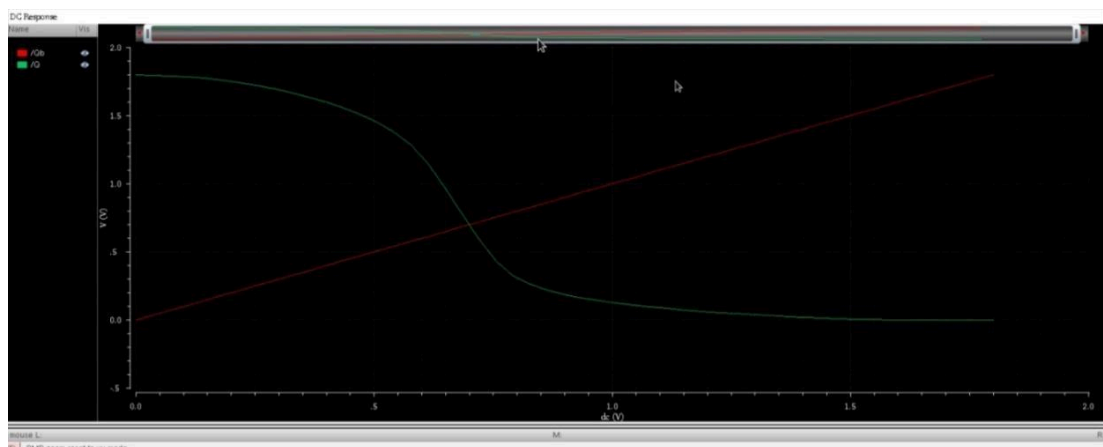
6. DC Analysis & Results

DC Analysis and SNM Explanation (Deep Content)

DC analysis is essential for evaluating the stability of the SRAM cell. It involves analyzing the voltage transfer characteristics (VTC) of the cross-coupled inverters. By plotting the VTC of one inverter and mirroring the VTC of the other inverter, a butterfly-shaped curve is obtained.

The largest square that can be fitted inside this butterfly curve represents the Static Noise Margin (SNM). SNM is a critical metric that quantifies the robustness of the SRAM cell against noise. A larger SNM indicates better stability and higher immunity to disturbances.

During the read operation, the SNM typically reduces due to the voltage division effect caused by the access transistors. Therefore, read SNM is usually lower than hold SNM, making it a critical design concern.



Factors affecting SNM include:

Transistor sizing

Supply voltage

Temperature

Process

variations

7. Conclusion

The design and analysis of the 6T SRAM cell presented in this report demonstrate the fundamental importance of memory cell optimization in modern VLSI systems. The 6T SRAM architecture continues to be the most widely used configuration in high-speed memory applications due to its simplicity, reliability, and efficient balance between performance and area. Through this study, a comprehensive understanding of the internal operation, design constraints, and performance trade-offs of the SRAM cell has been achieved.

The implementation of the 6T SRAM cell using CMOS technology highlights the critical role of transistor sizing in ensuring correct functionality. The strength relationship between pull-up, pull-down, and access transistors directly influences key parameters such as read stability, write ability, and data retention. Improper sizing can lead to issues such as read disturb, write failure, or reduced noise margins, which can compromise the reliability of the memory cell.

The simulation results obtained through DC and transient analyses provide valuable insights into the behavior of the SRAM cell under different operating conditions. The DC analysis, particularly the butterfly curve, plays a crucial role in evaluating the Static Noise Margin (SNM), which serves as a key indicator of cell stability. A higher SNM reflects better immunity to noise and process variations, making it an essential parameter in SRAM design. The transient analysis further illustrates the dynamic performance of the cell, including read and write delays, as well as switching characteristics.

One of the key observations from this work is the inherent trade-off between stability, speed, and power consumption. Enhancing read stability often requires stronger pull-down transistors, while improving write ability demands stronger access transistors. Similarly, reducing power consumption may impact performance metrics such as delay. Therefore, achieving an optimal design requires careful balancing of these conflicting requirements.

In addition, the study highlights the challenges associated with technology scaling. As device dimensions shrink, issues such as leakage current, reduced supply voltage, and increased process variability become more prominent. These factors significantly affect the reliability and performance of SRAM cells, necessitating advanced design techniques and optimization strategies.

Despite these challenges, the 6T SRAM cell remains a fundamental building block in modern integrated circuits, particularly in cache memory systems of processors and system-on-chip (SoC) designs. Its continued relevance is attributed to its robustness, fast operation, and compatibility with standard CMOS processes.

In conclusion, the 6T SRAM cell offers an effective solution for high-speed memory applications, provided that careful attention is given to design parameters and optimization techniques. The insights gained from this analysis can be utilized to further improve SRAM performance, especially in emerging technologies where power efficiency and reliability are critical.

